American Computer Science League

2025-2026 ● Contest 2: Short Problems Solutions ● Senior Division

1. Prefix/Infix/Postfix Notation

x x 1 + + x 2 + + x 3 + +
= (((x + (x+1)) + (x+2)) + (x+3))
x + (x+1) + (x+2) + (x+3) = 2026 can be solved for x.
4x + 6 = 2026
4x = 2020
2020 / 4 = 505
So 505 + 506 + 507 + 508 = 2026.

B. 505

2. Prefix/Infix/Postfix Notation

6 4 + 2 ^ 6 3 / / 8 4 / 4 8 / / +
(6 + 4) ^ 2 / (6 / 3) + (8 / 4) / (4 / 8))
= 10 ^ 2 / 2 + 2 / 0.5
= 100 / 2 + 4
= 50 + 4
= 54

C. 54

3. Bit-String Flicking

375_8 AND $A4_{16}$ OR 01110110_2
= 11111101 AND 10100100 OR 01110110
= 10100100 OR 01110110
= 11110110
= 11 110 110
= 366

D. 366

4. Bit-String Flicking

(NOT 01100) OR (LCIRC-1 01101 AND X) = X
(NOT 01100) OR (LCIRC-1 01101 AND abcde) = abcde
10011 OR 11010 AND abcde = abcde
10011 OR ab0d0 = abcde
1b011 = abcde
Therefore, a=1, b=*, c=0, d=1, e=1 so the bit string is 1*011.

A. 1*011

5. LISP (MULT 3 (SUB 8 6) (ADD 2 1 4) (DIV 6 2) (SQUARE 2)) = (MULT 3 2 7 3 4) = 42*12 = 504	E. 504
6. LISP V = '(r (b y) (o (p (g)))) (CAR (CDR (CAR (REVERSE (CDR V))))) = (CAR (CDR (CAR (REVERSE (CDR ‘(r (b y) (o (p (g))))))))) = (CAR (CDR (CAR (REVERSE '((b y) (o (p (g)))))))) = (CAR (CDR (CAR '((o (p (g))) (b y))))) = (CAR (CDR '(o (p (g))))) = (CAR '((p (g)))) = (p (g))	D. (p (g))